

Lorentzian propinquity convergence on truncated $SU(2) \times U(1)_T$ Krein spectral triples

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Abstract

We construct a Lorentzian analog of the Latrémolière quantum Gromov–Hausdorff propinquity, restricted to the Krein-positive state space, and prove the corresponding convergence theorem on truncated Krein spectral triples over the compact-temporal Wick-rotated manifold $S^3 \times S_T^1$. Concretely, let $\mathcal{K}_{n_{\max}, N_t} := \mathcal{H}_{\text{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ be the chirality-doubled Camporesi–Higuchi spinor space tensored with the N_t -mode Fourier truncation of $L^2(S_T^1)$, with fundamental symmetry $J_L = J_L^{\text{spat}} \otimes I_{N_t}$ at the Bizi–Brouder–Besnard [3] classification $(m, n) = (4, 6)$ in the Peskin–Schroeder chiral basis, and let $D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t})$ be the Lorentzian Dirac per van den Dungen 2016 [32] Proposition 4.1, with ∂_t Fourier-diagonal anti-Hermitian. Let $\mathcal{O}_{n_{\max}, N_t, T}^L$ be the truncated Lorentzian operator system built from chirality-doubled scalar 3- Y spatial multipliers tensored with the momentum-diagonal temporal multiplier subalgebra. On the Krein-positive subspace $\mathcal{K}^+ := \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$ we obtain a standard metric spectral triple $\mathcal{T}_{n_{\max}, N_t, T}^+$ to which Latrémolière 2017/2023 [16] machinery applies verbatim, and define the $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity $\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) := \Lambda(\mathcal{T}_1^+, \mathcal{T}_2^+)$. The main theorem reads

$$\Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0,$$

with $C_3^{\text{joint}} \leq 1$ asymptotically tight (inherited from Paper 38 [36] Lemma L3 via a structural identity for the joint commutator), and $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ the joint mass-concentration moment ($SU(2)$ factor inherits the $4/\pi$ asymptote of Paper 38 Lemma L2; $U(1)$ factor inherits the standard Fejér-on-the-circle rate). The proof is a five-lemma argument L1', L2, L3, L4, L5 transferred from Paper 38 with one substantive structural simplification: in the momentum-polynomial convention for temporal multipliers, the joint commutator $[D_L, a_s \otimes a_t]$ reduces *bit-exactly* to $i[D_{\text{GV}}, a_s] \otimes a_t$, so the temporal direction contributes nothing to the joint Lipschitz comparison; the joint C_3^{joint} equals the spatial $C_3^{SU(2)}$ verbatim. The Riemannian limit at $N_t = 1$ reduces Λ^L to Paper 38's bound bit-exactly (load-bearing falsifier). To our knowledge this is the first Lorentzian propinquity convergence theorem on truncated Krein spectral triples in the published math.OA literature. Honest scope: the construction is a *weak-form* propinquity on the $\mathcal{K}^+$ Hilbert-space restriction; a *strong-form* Latrémolière-style metric on Krein-signature spectral triples without $\mathcal{K}^+$ restriction remains an open NCG-math problem. The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] lives on a categorically different mathematical object (pre-length spaces with causal diamonds, not operator-algebraic spectral triples).

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1 Introduction

This paper closes the convergence-theorem leg of the Lorentzian-extension program for truncated GeoVac spectral triples, following the math.OA quartet Papers 38–40 [36–38] (Riemannian propinquity convergence on $SU(2)$, on tensor products of $SU(2)$ triples at distinct focal lengths, and on general compact Lie groups with bi-invariant metric) and the Lorentzian extensions Papers 42–44 [39–41] (Tomita–Takesaki modular structure on Riemannian truncations; four-witness Wick-rotation at the operator-system level on the Lorentzian Krein wedge; operator-system substrate of the Lorentzian Krein spectral triple at finite cutoff).

The dominant blocker for transferring Latrémolière propinquity [16, 17] to the Lorentzian setting is the absence of a published metric on Krein-signature spectral triples in their own right: Latrémolière’s construction crucially uses the operator-norm Lipschitz seminorm via the Dirac commutator $\|[D, \cdot]\|_{\text{op}}$, well-defined on Hilbert space but requiring an indefinite-inner-product analog on a Krein space (with the appropriate Krein-self-adjointness of D_L per van den Dungen 2016 [32] / BBB [3]). We sidestep this gap by working on the Krein-positive subspace $\mathcal{K}^+$, where the Krein product reduces to a positive-definite Hilbert inner product, and the standard Latrémolière machinery applies verbatim. We call this the *$\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity*.

1.1 What this paper does

We work on the compact-temporal Wick-rotated geometry $S^3 \times S_T^1$, with Riemannian metric $ds^2 = d\chi^2 + \sin^2 \chi d\Omega^2 + d\tau^2$ at canonical circumference $T = 2\pi$. The Krein structure J_L is preserved as an *algebraic* object (operator system plus symmetry, in the BBB (4, 6) classification) but no Lorentzian-causal-structure claim is made; the closed timelike-curve obstruction (Geroch [11]) is acknowledged explicitly via the Wick-rotated convention. The construction proceeds as a five-lemma argument transferred from Paper 38:

- (L1′) The Lorentzian operator-system substrate $\mathcal{O}_{n_{\max}, N_t, T}^L$ of Paper 44 [41], with propagation number 2 on the achievable envelope and trivial Krein-positivity at the operator-multiplier level.
- (L2) A joint central spectral Fejér kernel $K^{\text{joint}} = K_{n_{\max}}^{\text{SU}(2)} \otimes K_{N_t, T}^{U(1)}$ on $SU(2) \times U(1)$ with factorized Plancherel symbol and joint cb-norm $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max} + 1)$, via the Bożejko–Fendler central-multiplier equality [4, 26] on the amenable compact group product.
- (L3) The joint Lichnerowicz / Lipschitz comparison $\|[D_L, a]\|_{\text{op}} \leq C_3^{\text{joint}} \|\nabla^{\text{joint}} a\|_{\infty}$ with $C_3^{\text{joint}} \leq 1$, via the structural identity $[D_L, a_s \otimes a_t] = i[D_{\text{GV}}, a_s] \otimes a_t$ which makes the temporal direction contribute nothing to the joint commutator.
- (L4) A joint Berezin reconstruction $B_{n_{\max}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{n_{\max}, N_t, T}^L$ with positivity, contractivity, approximate identity (rate $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$), and L3 compatibility.
- (L5) Assembly of the propinquity tunneling pair $(B^{\text{joint}}, P^{\text{joint}})$ in the Latrémolière weak-form framework on $\mathcal{K}^+$, giving the main bound $\Lambda^L \leq C_3^{\text{joint}} \cdot \gamma^{\text{joint}} \rightarrow 0$.

1.2 Honest scope and named gaps

The construction is at the $\mathcal{K}^+$ -**restricted weak-form** level. The four named gaps to the strong-form construction are:

- (G1) *Strong-form Lorentzian propinquity*. A Latrémolière-style metric on Krein-signature spectral triples without $\mathcal{K}^+$ restriction is not in the published literature as of May 2026. The required ingredient is an indefinite-inner-product analog of the operator-norm Lipschitz seminorm with appropriate behavior under Krein-self-adjoint D_L . This is a multi-month NCG-math problem not addressed here.
- (G2) *De-compactification limit* $T \rightarrow \infty$. Our construction is at compact temporal radius T canonical $T = 2\pi$ (the BW modular period). The non-compact $\mathbb{R}_t$ limit re-introduces the joint compact $\times$ non-compact issue and is a separate program (Sprint L3c in our internal taxonomy).
- (G3) *Cross-manifold extensions* of the form $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ are blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (four-layer Coulomb/HO asymmetry from Paper 24 [34] § V) and are not addressed.
- (G4) *Inner-factor calibration data* (the Higgs / Yukawa selection question of the Connes–Marcolli SM construction) is orthogonal to the present convergence theorem.

1.3 Related work

Latrémolière’s quantum Gromov–Hausdorff propinquity [17] (Trans. AMS 2018) and the metric-spectral-triple weak-form extension [16] (Adv. Math. 2023) supply the convergence framework we work within. The most recent Riemannian-side continuity result is Latrémolière 2025 [9], treating analytic paths of Riemannian metrics; none of these constructions extend to Krein signature.

Connes–van Suijlekom 2021 [6] establishes the Gromov–Hausdorff convergence of spectral truncations on the circle S^1 and explicitly defers convergence on other manifolds (“elsewhere” on three occasions). The subsequent flat-structure program [8, 13, 14, 18, 30] covers $\mathbb{T}^d$, polynomial-growth groups, and the Berezin–Toeplitz S^2 , with universal $4/\pi$ asymptote on compact Lie groups established in Paper 40 [38] (also adapted to $SU(2)$ in Paper 38 [36]). All of these are strictly Riemannian.

The Lorentzian Krein-spectral-triple framework derives from Strohmaier [28] (axiomatic foundation), Franco–Eckstein [10] (algebraic causality), van den Dungen 2016 [32] (Krein spectral triples and fermionic action; the Wick-rotation lift used here), Bizi–Brouder–Besnard 2018 [3] ((m, n) classification of indefinite signatures used at (4, 6) West-coast in the present paper), Devastato–Lizzi–Martinetti [7], and the recent $SU(1,1)$ construction [33]; none of these works define a metric on the algebra of operator systems extracted from a Krein triple. The closest algebraic alternative is the Nieuviarts twist morphism [22, 23], which produces pseudo-Riemannian spectral triples from Riemannian ones via a twist on twisted spectral triples; this is not a metric and moreover (as noted in Paper 42 § 3.2) is restricted to even-dimensional manifolds, so does not apply to S^3 at KO-dimension 3.

The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] extends their earlier work [19] to Lorentzian pre-length spaces via ϵ -nets of causal diamonds; a downstream pure-metric extension via the intrinsic timed-Hausdorff distance appears in Che–Perales–Sormani [21]. This is conceptually adjacent (all are “Lorentzian Gromov–Hausdorff-type convergence”) but lives on a categorically different mathematical object: pre-length spaces with causal-diamond / timed-Hausdorff geometry, not operator-algebraic spectral triples. The two families of constructions are not in competition; neither implies the other.

1.4 Outline

§ 2 fixes conventions: Krein space, Lorentzian Dirac, operator system, joint Fejér kernel, Krein-positive subspace, and the definition of the $\mathcal{K}^+$ -restricted weak-form propinquity. § 3 reviews the L1' operator-system substrate (Paper 44). § 4 proves the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3). § 5 states and proves the main theorem (L5 propinquity assembly). § 6 records the numerical verification panel and the bit-exact Riemannian-limit recovery at $N_t = 1$. § 7 discusses the four-witness Wick-rotation connection (Paper 42/43), records the open questions (strong-form, de-compactification, cross-manifold), and positions the work against Mondino–Sämman. Appendix A records the symbolic Plancherel calculation for the joint Fejér symbol. Appendix B records the numerical panel data.

2 Preliminaries

2.1 Compact-temporal manifold and Wick-rotated metric

The compact-temporal Lorentzian / Wick-rotated manifold is $S^3 \times S_T^1$, with $S^3 = \text{SU}(2)$ the round unit three-sphere (spatial slot) and $S_T^1 = \mathbb{R}/T\mathbb{Z}$ the temporal circle of circumference T . The Riemannian / Wick-rotated metric is

$$ds^2 = d\chi^2 + \sin^2 \chi d\Omega_2^2 + d\tau^2, \quad (1)$$

with $\chi \in [0, \pi]$ the polar angle on S^3 , $d\Omega_2^2$ the round metric on the $\mathbb{S}^2$ base of the Hopf fibration, and $\tau \in \mathbb{R}/T\mathbb{Z}$ the temporal coordinate. We treat the Krein structure J_L defined below as an *algebraic* object that encodes the BBB (4,6) chirality grading without claiming a Lorentzian-causal interpretation of the underlying metric: Geroch's no-CTC obstruction [11] prevents a Lorentzian metric on $S^3 \times S_T^1$ in which S_T^1 is timelike, so the present construction operates on the Wick-rotated geometry and lifts the Krein structure as indefinite-inner-product datum (per Strohmaier [28], vdD [32], BBB [3]).

The canonical choice $T = 2\pi$ matches the Bisognano–Wichmann [2] modular period at unit surface gravity; this is consistent with the four-witness arc closure of Paper 43 [40]. The proof of the main theorem holds for any $T > 0$; explicit dependence on T enters the $U(1)$ factor of the joint mass-concentration moment as $O(T/N_t)$.

2.2 Krein space and fundamental symmetry

Per Paper 43 [40]§ 2 / Paper 44 [41]§ 2, the Krein space is

$$\mathcal{K}_{n_{\max}, N_t} := \mathcal{H}_{\text{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}, \quad (2)$$

where $\mathcal{H}_{\text{GV}}^{n_{\max}}$ is the chirality-doubled Camporesi–Higuchi [5] spinor Hilbert space truncated at the $n_{\max}$ -th Fock shell, of dimension

$$\dim \mathcal{H}_{\text{GV}}^{n_{\max}} = \frac{2}{3} n_{\max}(n_{\max} + 1)(n_{\max} + 2), \quad (3)$$

and $\mathbb{C}^{N_t}$ is the N_t -mode Fourier truncation of $L^2(S_T^1)$. The fundamental symmetry is

$$J_L = J_L^{\text{spat}} \otimes I_{N_t}, \quad (4)$$

with $J_L^{\text{spat}} = \gamma^0$ the chirality-swap in the Peskin–Schroeder chiral basis, $(J_L^{\text{spat}})^2 = +I$.

The BBB (m, n) classification at signature $(s, t) = (3, 1)$ West-coast assigns $(m, n) = (s + t, t - s \bmod 8) = (4, 6)$ via BBB [3] Table 3, fixing $\epsilon = +1, \epsilon'' = -1, \kappa = -1, \kappa'' = +1$. These signs were verified bit-exact at finite cutoff in Paper 43 § 5 [40].

2.3 Lorentzian Dirac operator

The Lorentzian Dirac, per vdD 2016 [32] Prop. 4.1 applied at signature (3,1), reads

$$D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t}), \quad (5)$$

with ∂_t Fourier-diagonal anti-Hermitian on $\mathbb{C}^{N_t}$: in the momentum eigenbasis $\{|k\rangle : k \in \mathbb{Z}, |k| \leq K_{\text{max}}\}$ with $K_{\text{max}} := \lfloor (N_t - 1)/2 \rfloor$ for odd N_t or $N_t/2$ for even N_t ,

$$\partial_t = i \text{diag}(\omega_k), \quad \omega_k = 2\pi k/T, \quad (6)$$

so $\partial_t^\dagger = -\partial_t$. The spatial part D_{GV} is the truthful Camporesi–Higuchi spatial Dirac on $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$, diagonal in the spinor basis with eigenvalue $\chi(n+1/2)$ where $\chi \in \{+1, -1\}$ is the spinor chirality and n the Fock-shell index. Krein-self-adjointness $D_L^\times := J_L D_L^\dagger J_L = D_L$ is verified by direct computation: $\gamma^{0\dagger} = \gamma^0$ (Hermitian chirality-swap), $\{\gamma^0, D_{\text{GV}}\} = 0$ in the chiral basis, $\partial_t^\dagger = -\partial_t$, and $J_L^2 = +I$ combine to give the stated identity. The sign $i^t = +i$ in (5) is forced by this requirement (an opposite sign would give an anti-Krein-self-adjoint D_L); see Paper 43 § 3 [40] for the explicit derivation.

2.4 Truncated Lorentzian operator system

Per Paper 44 [41], the truncated Lorentzian operator system is the pure-tensor span

$$\mathcal{O}_{n_{\text{max}}, N_t, T}^L := \text{span}_{\mathbb{C}}\{M_{N, L, M}^{\text{spat}} \otimes M_p^{\text{temp}} : (N, L, M) \in \mathcal{I}_{n_{\text{max}}}^{\text{spat}}, p \in \{0, 1, \dots, N_t - 1\}\}, \quad (7)$$

where each spatial multiplier $M_{N, L, M}^{\text{spat}} = \text{blkdiag}(W_{N, L, M}, W_{N, L, M})$ is the chirality-doubled lift of the Avery–Wen–Avery Weyl-sector multiplier $W_{N, L, M}$ (the Clebsch–Gordan / Avery 3- Y integral construction of Paper 32 [35] § III), and the temporal multipliers M_p^{temp} are momentum-polynomial diagonal:

$$M_p^{\text{temp}} = \text{diag}(\omega_k^p)_{|k| \leq K_{\text{max}}}. \quad (8)$$

Vandermonde invertibility at N_t distinct frequencies ω_k ensures that the span (7) coincides with the span of Fourier-mode indicators $M_q^{\text{temp}} = \text{diag}(\delta_{k, q})$, and we use whichever basis is more convenient.

By Paper 44 § 3, $\mathcal{O}_{n_{\text{max}}, N_t, T}^L$ is a $*$ -closed linear subspace of $\mathcal{B}(\mathcal{K}_{n_{\text{max}}, N_t})$, contains the identity, and is *not* multiplicatively closed (a witness pair is constructed in Paper 44 § 5). The propagation number is $\text{prop}(\mathcal{O}_{n_{\text{max}}, N_t}^L) = 2$ on the Weyl-doubled achievable envelope, matching the Riemannian Paper 32 § III $\text{prop}=2$ verbatim; $\text{prop} = \infty$ on the full $\mathcal{B}(\mathcal{K})$ envelope due to the chirality-doubling and the commutative temporal subalgebra.

2.5 Joint central spectral Fejér kernel

The $\text{SU}(2)$ factor is the Paper 38 [36] L2-DEF central spectral Fejér kernel

$$K_{n_{\text{max}}}^{\text{SU}(2)}(g) := \frac{1}{Z_{n_{\text{max}}}^{\text{SU}(2)}} \left| \sum_{j \leq j_{\text{max}}} \sqrt{2j+1} \chi_j(g) \right|^2, \quad (9)$$

with $j_{\text{max}} = (n_{\text{max}} - 1)/2$ and $Z_{n_{\text{max}}}^{\text{SU}(2)} = n_{\text{max}}(n_{\text{max}} + 1)/2$, central Plancherel symbol

$$\hat{K}_{n_{\text{max}}}^{\text{SU}(2)}(j) = \frac{2j+1}{Z_{n_{\text{max}}}^{\text{SU}(2)}} \mathbf{1}_{\{j \leq j_{\text{max}}\}}. \quad (10)$$

The $U(1)$ factor is the standard Fejér kernel on S_T^1 ,

$$K_{N_t, T}^{U(1)}(\theta) := \frac{1}{N_t} \left| \sum_{|k| \leq K_{\text{max}}} e^{i 2\pi k \theta / T} \right|^2, \quad (11)$$

with Plancherel symbol

$$\hat{K}_{N_t}^{U(1)}(k) = \max\left(0, \frac{N_t+1-2|k|}{N_t+1}\right). \quad (12)$$

The joint kernel is the tensor product

$$K_{n_{\max}, N_t, T}^{\text{joint}}(g, \theta) := K_{n_{\max}}^{\text{SU}(2)}(g) \cdot K_{N_t, T}^{U(1)}(\theta), \quad (13)$$

with factorized Plancherel symbol

$$K^{\hat{\text{joint}}}(j, k) = \hat{K}_{n_{\max}}^{\text{SU}(2)}(j) \cdot \hat{K}_{N_t}^{U(1)}(k). \quad (14)$$

Both factors are non-negative, normalized (L^1 norm = 1), and central (class functions on their respective groups); the tensor product inherits these properties on $\text{SU}(2) \times U(1)$.

2.6 Krein-positive subspace and weak-form propinquity

The Krein-positive subspace is

$$\mathcal{K}^+ := \{|\psi\rangle \in \mathcal{K} : J_L|\psi\rangle = +|\psi\rangle\}, \quad (15)$$

the $+1$ -eigenspace of J_L , of dimension $\dim \mathcal{K}^+ = \dim \mathcal{K}/2 = \frac{1}{3} n_{\max}(n_{\max}+1)(n_{\max}+2)N_t$. The Krein inner product $\langle \psi, \phi \rangle_{J_L} := \langle \psi, J_L \phi \rangle$ restricted to $\mathcal{K}^+$ reduces to the standard Hilbert inner product $\langle \psi, \phi \rangle$, since $J_L|\phi\rangle = |\phi\rangle$ on $\mathcal{K}^+$. Hence $\mathcal{K}^+$ is a genuine Hilbert space.

The orthogonal projection onto $\mathcal{K}^+$ is $P_+ := (I + J_L)/2$ (and dually $P_- := (I - J_L)/2$ for the -1 -eigenspace).

Definition 2.1 ($\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity). For two Lorentzian truncated Krein metric spectral triples $\mathcal{T}_i^L = (\mathcal{O}_i^L, \mathcal{K}_i, D_i^L, J_i^L)$ with the construction above, the $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity is

$$\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) := \Lambda(\mathcal{T}_1^+, \mathcal{T}_2^+), \quad (16)$$

where $\mathcal{T}_i^+ := (P_i^+ \mathcal{O}_i^L P_i^+, \mathcal{K}_i^+, P_i^+ D_i^L P_i^+)$ is the $\mathcal{K}^+$ -restricted standard metric spectral triple and Λ is the standard Latrémolière metric-spectral-triple propinquity [16].

The construction is well-defined precisely because every Berezin-image operator and every truncation projection used below commutes with J_L at the operator level (Paper 44 § 4 / Sub-sprint C § 6 of the internal memos), so $\mathcal{K}^+$ is preserved and the standard machinery applies.

2.7 Joint gradient norms

For a pure-tensor symbol $f = f_s \otimes f_t$ on $S^3 \times S_T^1$, the L^1 -additive and L^2 -Pythagorean joint gradient norms are

$$\|\nabla^{\text{joint}, L^1} f\|_\infty := \|\nabla_x f_s\|_\infty \cdot \|f_t\|_\infty + \|f_s\|_\infty \cdot \|\partial_t f_t\|_\infty, \quad (17)$$

$$\|\nabla^{\text{joint}, L^2} f\|_\infty := \sup_{(x,t)} \sqrt{|\nabla_x f(x, t)|^2 + |\partial_t f(x, t)|^2}. \quad (18)$$

For general f , both extend by linearity over the Peter–Weyl $\times$ Fourier expansion and obey the standard triangle inequalities. The L^2 form is dominated by the L^1 form via $\sqrt{a^2 + b^2} \leq a + b$ for $a, b \geq 0$; both forms upper-bound the same Lipschitz seminorm with the same constant in the proof of Lemma 4.4.

3 Lemma L1': the operator-system substrate

Lemma L1' supplies the truncated metric spectral triple on which the four subsequent lemmas operate. It is the closure result of Paper 44 [41], summarized here for completeness.

Lemma 3.1 (L1', Paper 44). *The collection $\mathcal{T}_{n_{\max}, N_t, T}^L := (\mathcal{O}_{n_{\max}, N_t, T}^L, \mathcal{K}_{n_{\max}, N_t}, D_{n_{\max}, N_t, T}^L, J_{n_{\max}, N_t}^L)$ defined in § 2.2–2.4 is a Lorentzian truncated Krein metric spectral triple in the sense of vdD [32] / BBB [3]:*

- (a) $\mathcal{O}^L$ is a unital $*$ -closed linear subspace of $\mathcal{B}(\mathcal{K})$.
- (b) D_L is Krein-self-adjoint, $D_L^\times = D_L$.
- (c) J_L is a fundamental symmetry, $J_L^2 = +I$, $J_L^\dagger = J_L$.
- (d) The propagation number $\text{prop}(\mathcal{O}^L) = 2$ on the Weyl-doubled achievable envelope.
- (e) Every multiplier in $\mathcal{O}^L$ commutes with J_L : $[J_L, a] = 0$ bit-exact for every $a \in \mathcal{O}^L$.
- (f) At $N_t = 1$, the construction reduces bit-exactly to the Riemannian chirality-doubled truncated spectral triple on $\mathcal{H}_{\text{GV}}^{n_{\max}}$ (Frobenius residual 0.0 in float64).

Proof. (a)–(c) are direct from the construction and Paper 43 § 2,3,5. (d) is Paper 44 § 3 (via the structural identity $\dim((\mathcal{O}^L)^2) = \dim_{\text{Weyl}}^2 N_t$ on the Weyl-doubled envelope). (e) follows from $[\gamma^0, M^{\text{spat}}] = 0$ (chirality-swap acts trivially on the diagonal block-doubled spatial multiplier) and $[I_{N_t}, M^{\text{temp}}] = 0$ (trivially). (f) is Paper 44 § 2 (load-bearing falsifier F1). $\square$

Remark 3.2 (Krein-positivity at the operator-multiplier level is trivial). By Lemma 3.1(e), every operator $a \in \mathcal{O}^L$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$. Consequently, the $\mathcal{K}^+$ -restricted operator system $\mathcal{O}^{L,+} := P_+ \mathcal{O}^L P_+$ has the same propagation number 2 as the unrestricted system, and the standard Latrémolière metric-spectral-triple machinery applies on $(\mathcal{O}^{L,+}, \mathcal{K}^+, P_+ D_L P_+)$ verbatim. The non-trivial Krein-positive structure shifts to the *state* level (Wasserstein–Kantorovich on $\mathcal{K}^+$ states), which is the natural setting for the L5 propinquity assembly in § 5.

4 The three analytical lemmas

We now prove the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3) on the substrate supplied by Lemma 3.1. The mechanisms transfer from Paper 38 / Paper 40 with one substantive simplification per lemma, arising from the pure-tensor structure of $\mathcal{O}^L$ and the momentum-diagonal structure of the temporal multipliers.

4.1 Lemma L2: joint cb-norm of the central Schur multiplier

Lemma 4.1 (L2, joint cb-norm). *For all $n_{\max} \geq 1$ and $N_t \geq 1$, the central Schur multiplier $S_{K^{\text{joint}}}$ on the central subalgebra $Z(C(\text{SU}(2) \times U(1)))$ has cb-norm*

$$\|S_{K^{\text{joint}}}_{n_{\max}, N_t, T}\|_{\text{cb}} = \|S_{K^{\text{SU}(2)}}\|_{\text{cb}} \cdot \|S_{K^{U(1)}}\|_{\text{cb}} = \frac{2}{n_{\max} + 1} \cdot 1 = \frac{2}{n_{\max} + 1}. \quad (19)$$

Proof. The compact group $G = \text{SU}(2) \times U(1)$ is amenable (direct product of two compact groups, both amenable by Haar invariance [1, 25]). By the Bożejko–Fendler central-multiplier equality [4, 26] Ch. 8 Thm. 8.10, every central Fourier multiplier symbol $m \in \ell^\infty(\hat{G})$ has Schur cb-norm equal to its ℓ^∞ norm.

Factor 1 (SU(2)). Inherited from Paper 38 [36] Lemma L2(c) verbatim: the Plancherel symbol $\hat{K}_{n_{\max}}^{\text{SU}(2)}(j) = (2j+1)/Z_{n_{\max}}^{\text{SU}(2)}$ is monotonically increasing in $j \in \{0, \frac{1}{2}, 1, \dots, j_{\max}\}$, so its supremum is $\hat{K}_{n_{\max}}^{\text{SU}(2)}(j_{\max}) = n_{\max}/Z_{n_{\max}}^{\text{SU}(2)} = 2/(n_{\max}+1)$.

Factor 2 (U(1)). $U(1)$ is abelian, so all irreps are 1-dimensional and every multiplier symbol is automatically central. The Fejér symbol (12) is non-negative, supported on $|k| \leq K_{\max}$, and monotonically decreasing in $|k|$ from $\hat{K}_{N_t}^{U(1)}(0) = 1$, so its supremum is 1, attained at $k = 0$.

Joint cb-norm via factorization. The central subalgebra of $C(\text{SU}(2) \times U(1))$ factorizes as the spatial C^* -tensor product $Z(C(\text{SU}(2))) \hat{\otimes} Z(C(U(1)))$ (Takesaki [29] Thm. IV.4.14, commutative-factor case). The joint Plancherel symbol (14) factorizes correspondingly. Both factor symbols are non-negative, so

$$\sup_{(j,k)} \hat{K}^{\text{SU}(2)}(j) \hat{K}^{U(1)}(k) = \left(\sup_j \hat{K}^{\text{SU}(2)}(j) \right) \left(\sup_k \hat{K}^{U(1)}(k) \right).$$

Applying Bożejko–Fendler in reverse to each factor (and once jointly on the amenable product) gives (19). $\square$

Remark 4.2 (N_t -independence of the cb-norm). The joint cb-norm $2/(n_{\max}+1)$ is independent of N_t : the $U(1)$ factor caps at 1 regardless of N_t . The N_t -dependence of the joint convergence rate enters the *mass-concentration moment* γ^{joint} in Lemma 4.8, not the cb-norm. This is the structurally correct behavior: the cb-norm controls the Lipschitz-distortion height in the propinquity assembly (§ 5), while γ^{joint} controls the reach. They are different invariants of the same kernel.

Remark 4.3 ($4/\pi$ asymptote on the SU(2) factor). Paper 38 § 3.2 (with the universal Plancherel- $\times$ -Vandermonde cancellation theorem from Paper 40 § 3.2 [38]) gives the asymptotic rate constant

$$\lim_{n_{\max} \rightarrow \infty} \frac{n_{\max} \cdot \gamma_{n_{\max}}^{\text{SU}(2)}}{\log n_{\max}} = \frac{4}{\pi}.$$

This is the universal compact-Lie-group rate constant on the bi-invariant metric and is independent of rank, Weyl-group order, and Lie type within the class. The $U(1)$ factor inherits a separate Fejér-on-the-circle rate.

4.2 Lemma L3: joint Lichnerowicz / Lipschitz comparison

Lemma 4.4 (L3, joint Lichnerowicz). *For every $a \in \mathcal{O}_{n_{\max}, N_t, T}^L$ and every $(n_{\max}, N_t, T)$,*

$$\| [D_L, a] \|_{\text{op}} \leq C_3^{\text{joint}}(n_{\max}, N_t) \cdot \|\nabla^{\text{joint}} a\|_{\infty}, \quad (20)$$

where ∇^{joint} is either the L^1 -additive or L^2 -Pythagorean joint gradient norm (17), (18). The constant satisfies

$$C_3^{\text{joint}}(n_{\max}, N_t) \leq C_3^{\text{SU}(2)}(n_{\max}) \leq \sup_{2 \leq N \leq n_{\max}} \frac{N-1}{\sqrt{N^2-1}} \xrightarrow{n_{\max} \rightarrow \infty} 1^-, \quad (21)$$

inherited verbatim from Paper 38 [36] Lemma L3 (and is asymptotically tight on the natural panel).

Proof. We first establish the structural identity: for every pure-tensor $a = a_s \otimes a_t \in \mathcal{O}^L$,

$$[D_L, a_s \otimes a_t] = i [D_{\text{GV}}, a_s] \otimes a_t \quad (\text{bit-exact}). \quad (22)$$

Expand $[D_L, a_s \otimes a_t] = i [\gamma^0 \otimes \partial_t, a_s \otimes a_t] + i [D_{\text{GV}} \otimes I_{N_t}, a_s \otimes a_t]$ and analyze each term.

Term A (vanishing cross term). Since $a_s = \text{blkdiag}(W, W)$ has identical diagonal blocks in chirality and γ^0 is the chirality-swap acting as $\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$ on the chirality indices,

$$\gamma^0 a_s = \begin{pmatrix} 0 & W \\ W & 0 \end{pmatrix} = a_s \gamma^0, \quad [\gamma^0, a_s] = 0.$$

Furthermore, a_t and ∂_t are both diagonal in the momentum basis (the temporal multiplier algebra is the commutative diagonal subalgebra of $M_{N_t}(\mathbb{C})$), so $[\partial_t, a_t] = 0$. The cross term therefore vanishes bit-exactly: $[\gamma^0 \otimes \partial_t, a_s \otimes a_t] = 0$.

Term B (purely spatial). $[D_{\text{GV}} \otimes I_{N_t}, a_s \otimes a_t] = [D_{\text{GV}}, a_s] \otimes a_t$.

Combining gives (22). By linearity over the operator-system span, the same identity extends to every $a \in \mathcal{O}^L$ written as a finite linear combination of pure-tensor generators.

Tensor-product operator-norm factorization $\|A \otimes B\|_{\text{op}} = \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}$ and Paper 38 Lemma L3 on the spatial factor give, for any pure-tensor $a = a_s \otimes a_t$,

$$\|[D_L, a]\|_{\text{op}} = \|[D_{\text{GV}}, a_s]\|_{\text{op}} \cdot \|a_t\|_{\text{op}} \leq C_3^{\text{SU}(2)}(n_{\text{max}}) \cdot \|\nabla_x f_s\|_{\infty} \cdot \|a_t\|_{\text{op}}, \quad (23)$$

where f_s is the Berezin pre-image of a_s . For momentum-diagonal a_t , $\|a_t\|_{\text{op}} = \|f_t\|_{\infty}$ (with f_t the corresponding Fourier symbol). Since $\|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} \leq \|\nabla^{\text{joint}, L^1} f\|_{\infty}$ by (17) (the second summand is non-negative), and similarly $\leq \|\nabla^{\text{joint}, L^2} f\|_{\infty}$ by $\sqrt{a^2 + b^2} \geq a$ for $a, b \geq 0$, we obtain (20) for pure-tensor a . The extension to general $a = \sum_j c_j(a_s^j \otimes a_t^j)$ follows by triangle inequality on both sides.

The bound (21) is Paper 38 § L3 verbatim; asymptotic tightness $\rightarrow 1^-$ is via the per-harmonic Avery–Wen–Avery saturation on $Y_{N,0,0}^{(3)}$ as $N \rightarrow n_{\text{max}}$. $\square$

Remark 4.5 (The temporal direction contributes zero). The identity (22) is the substantive structural content of Lemma 4.4. There is *no* cross term and *no* separate $U(1)$ Lichnerowicz lemma: the joint commutator is the spatial commutator times the temporal operator-norm, and the temporal slot enters the bound multiplicatively at value 1. This is stronger than the $C_3 \leq 1 + \text{cross estimate}$ one would expect from a generic tensor extension; the structural reason is that the temporal multiplier algebra is commutative diagonal and ∂_t is itself diagonal, so they commute bit-exactly. The Wick-rotated convention here is essential: a Lorentzian-causal temporal derivative would not be diagonal in the same basis.

Remark 4.6 (L^1 and L^2 give the same C_3). Both L^1 -additive (17) and L^2 -Pythagorean (18) joint gradient norms upper-bound the right-hand side of (23) by the same constant $C_3^{\text{SU}(2)}$: the temporal contribution to $\|[D_L, a]\|_{\text{op}}$ is zero (by (22)), so only the spatial part of the joint gradient matters in the inequality.

4.3 Lemma L4: joint Berezin reconstruction

Definition 4.7 (Joint Berezin reconstruction). The joint Berezin reconstruction map is $B_{n_{\text{max}}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{n_{\text{max}}, N_t, T}^L$ defined on the joint Peter–Weyl $\times$ Fourier expansion $f(\omega, t) = \sum_{N,L,M,q} c_{N,L,M,q} Y_{N,L,M}^{(3)}(\omega) e^{iqt/R_T}$ (with $R_T = T/(2\pi)$) by

$$B^{\text{joint}}(f) = \sum_{\substack{N \leq n_{\text{max}} \\ |q| \leq K_{\text{max}}}} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \hat{K}_{n_{\text{max}}}^{\text{SU}(2)}(N) \hat{K}_{N_t}^{U(1)}(q) c_{N,L,M,q} (M_{N,L,M}^{\text{spat}} \otimes M_q^{\text{temp}}). \quad (24)$$

Equivalently, $B^{\text{joint}}(f) = P^{\text{joint}}(K^{\text{joint}} * f)P^{\text{joint}}$ where $P^{\text{joint}} := P_{n_{\text{max}}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$ is the joint truncation projection onto the truncated subspace of $L^2(S^3 \times S_T^1)$ spanned by the kept Peter–Weyl $\times$ Fourier modes. The two definitions agree on the central subalgebra by joint Plancherel (14).

Lemma 4.8 (L4, joint Berezin properties). $B_{n_{\max}, N_t, T}^{\text{joint}}$ satisfies the following four properties.

- (a) Positivity. If $f \geq 0$ pointwise on $S^3 \times S_T^1$, then $B^{\text{joint}}(f) \geq 0$ as a positive-semidefinite Hermitian element of $M_{\dim \mathcal{K}}(\mathbb{C})$.
- (b) Contractivity. $\|B^{\text{joint}}\|_{\text{cb}} \leq \|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1) \leq 1$, and in particular $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|f\|_{\infty}$.
- (c) Approximate identity. For every $f \in C^\infty(S^3 \times S_T^1)$,

$$\|B^{\text{joint}}(f) - P^{\text{joint}} M_f P^{\text{joint}}\|_{\text{op}} \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_{\infty}, \quad (25)$$

with $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \rightarrow 0$ as $(n_{\max}, N_t) \rightarrow (\infty, \infty)$.

- (d) L3 compatibility. For every $f \in C^\infty(S^3 \times S_T^1)$,

$$\|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \leq C_3^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_{\infty}, \quad (26)$$

with the same $C_3^{\text{joint}} \leq 1$ as in Lemma 4.4.

- (e) Krein-positivity preservation. For every f , $[J_L, B^{\text{joint}}(f)] = 0$ bit-exact, hence $B^{\text{joint}}(f)$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$.

Proof. (a) *Positivity.* The convolution form $B^{\text{joint}}(f) = P^{\text{joint}} (K^{\text{joint}} * f) P^{\text{joint}}$ writes $B^{\text{joint}}(f)$ as a UCP compression of a non-negative function (since both factor kernels are non-negative pointwise: $K^{\text{SU}(2)} = |\cdot|^2 \cdot Z^{-1} \geq 0$ and $K^{U(1)} = |\cdot|^2 \cdot N_t^{-1} \geq 0$). The convolution of two non-negative integrands against the positive joint Haar measure is non-negative, and the UCP compression of a non-negative multiplication operator is positive-semidefinite.

(b) *Contractivity.* The convolution form gives $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|K^{\text{joint}} * f\|_{\infty} \leq \|K^{\text{joint}}\|_{L^1} \cdot \|f\|_{\infty}$ by Young's inequality at the sup endpoint, and $\|K^{\text{joint}}\|_{L^1} = 1$ (both factors are normalized probability densities; Haar on a product factorizes). The sharper cb-norm bound is via Lemma 4.1: B^{joint} is the composition of the UCP projection P^{joint} (cb-norm 1) with the central Schur multiplier $S_{K^{\text{joint}}}$ (cb-norm $2/(n_{\max}+1)$ by Lemma 4.1), giving $\|B^{\text{joint}}\|_{\text{cb}} \leq 2/(n_{\max}+1)$.

(c) *Approximate identity.* For a pure-tensor symbol $f = f_s \otimes f_t$, the convolution-deficit splits:

$$K^{\text{joint}} * f - f = (K^{\text{SU}(2)} * f_s - f_s) \otimes (K^{U(1)} * f_t) + f_s \otimes (K^{U(1)} * f_t - f_t).$$

By Paper 38 [36] Lemma L4(c) on the spatial factor, $\|K^{\text{SU}(2)} * f_s - f_s\|_{\infty} \leq \gamma_{n_{\max}}^{\text{SU}(2)} \cdot \|\nabla_x f_s\|_{\infty}$, with $\gamma_{n_{\max}}^{\text{SU}(2)} = O(\log n_{\max}/n_{\max})$ (Paper 38 § L2 quantitative rate, $4/\pi$ asymptote). By the standard Fejér-on-the-circle approximation rate (Katznelson [15] Ch. I), $\|K^{U(1)} * f_t - f_t\|_{\infty} \leq \gamma_{N_t, T}^{U(1)} \cdot \|\partial_t f_t\|_{\infty}$ with $\gamma^{U(1)} = O(T/N_t)$. Combining with the L^1 -additive triangle decomposition and $\|K^{U(1)} * f_t\|_{\infty} \leq \|f_t\|_{\infty}$ (Young) gives

$$\|K^{\text{joint}} * f - f\|_{\infty} \leq \max(\gamma_{n_{\max}}^{\text{SU}(2)}, \gamma_{N_t, T}^{U(1)}) \cdot \|\nabla^{\text{joint}, L^1} f\|_{\infty}.$$

We define $\gamma^{\text{joint}} := \max(\gamma^{\text{SU}(2)}, \gamma^{U(1)})$. Compression-contraction on L^2 does not increase the operator norm, so (25) follows. Linearity and the triangle inequality extend to general f .

(d) *L3 compatibility.* By linearity over (24), $[D_L, B^{\text{joint}}(f)] = \sum c_{N, L, M, q} K^{\hat{\text{joint}}}(N, q) i[D_{\text{GV}}, M_{N, L, M}^{\text{spat}}] \otimes M_q^{\text{temp}}$ by the L3 structural identity (22). Tensor-norm factorization and Paper 38 § L3 give $\|[D_{\text{GV}}, M_{N, L, M}^{\text{spat}}]\|_{\text{op}} \leq (N-1) \cdot \|M^{\text{spat}}\|_{\text{op}}$ with $C_3^{(N)} = (N-1)/\sqrt{N^2-1}$ on the natural Avery harmonic. Combining with $K^{\hat{\text{joint}}}(N, q) \leq 1$ and triangle inequality gives (26).

(e) *Krein-positivity preservation.* $[J_L, M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}}] = [J_L^{\text{spat}}, M^{\text{spat}}] \otimes M^{\text{temp}} + M^{\text{spat}} \otimes [I_{N_t}, M^{\text{temp}}] = 0$ by the same argument as Lemma 3.1(e), extended by linearity. $\square$

Remark 4.9 (Tensor-product factorization on pure tensors). For a pure-tensor $f = f_s \otimes f_t$, the joint Berezin factorizes:

$$B^{\text{joint}}(f_s \otimes f_t) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes B^{U(1)}(f_t), \quad (27)$$

where $B_{\chi\text{-d}}^{\text{SU}(2)}$ is the chirality-doubled spinor lift of Paper 38's spatial Berezin and $B^{U(1)}$ is the standard Fejér-Berezin on S_T^1 . This factorization makes Lemma 4.8(a)–(e) follow from the corresponding factor-wise properties of Paper 38 L4 (SU(2) side) and standard Fejér-on- S^1 approximation theory ($U(1)$ side).

5 Main theorem (Lemma L5): propinquity assembly

5.1 The tunneling pair

Definition 5.1 (Joint tunneling pair). The joint tunneling pair between the truncated triple $\mathcal{T}_{n_{\max}, N_t, T}^L$ and the continuum reference triple $\mathcal{T}_{S^3 \times S_T^1}^L$ is the pair $(B_{n_{\max}, N_t, T}^{\text{joint}}, P_{n_{\max}, N_t}^{\text{joint}})$, with B^{joint} the joint Berezin map of Definition 4.7 and $P^{\text{joint}} = P_{n_{\max}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$ the joint truncation projection on $L^2(S^3 \times S_T^1)$.

The continuum reference triple is $\mathcal{T}_{S^3 \times S_T^1}^L := (C^\infty(S^3 \times S_T^1), L^2(S^3, \Sigma_{\text{CH}}) \otimes L^2(S_T^1), D_\infty^L, J_\infty^L)$ with Σ_{CH} the chirality-doubled Camporesi–Higuchi spinor bundle and D_∞^L, J_∞^L the natural continuum analogs of D_L and J_L .

By Lemmas 3.1–4.8, both B^{joint} and P^{joint} are UCP (with the Berezin cb-norm bound $\|B^{\text{joint}}\|_{\text{cb}} \leq 2/(n_{\max} + 1) \leq 1$; the projection P^{joint} is a Stinespring orthogonal projection), and both commute with J_L at the operator level. Hence the $\mathcal{K}^+$ -restricted pair $(P_+ B^{\text{joint}}(\cdot) P_+, P_+ P^{\text{joint}}(\cdot) P_+)$ is a valid Latrémolière tunneling pair between the $\mathcal{K}^+$ -restricted truncated and continuum triples in Definition 2.1.

5.2 The four propinquity constituents

Following Latrémolière [16]§ 4 (metric-spectral-triple weak-form propinquity bound) and Paper 38 § L5, the propinquity bound for a direct UCP tunneling pair is

$$\Lambda(\mathcal{T}_{n_{\max}, N_t, T}^+, \mathcal{T}_{S^3 \times S_T^1}^+) \leq \max(\text{reach}_B, \text{reach}_P, \text{height}_B, \text{height}_P). \quad (28)$$

Proposition 5.2 (Reach and height bounds). *The four propinquity constituents satisfy:*

$$\text{reach}_B \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (29)$$

$$\text{reach}_P \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (30)$$

$$\text{height}_B \leq C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (31)$$

$$\text{height}_P = 0. \quad (32)$$

Proof. (29) is Lemma 4.8(c) (approximate identity), applied on the unit-Lipschitz ball $\{f : \|\nabla^{\text{joint}} f\|_\infty \leq 1\}$. The bound restricts unchanged to $\mathcal{K}^+$ since P_+ is UCP.

(30) is the dual roundtrip: start with M_f on the continuum side, apply P^{joint} to compress to $\mathcal{O}^L$, and lift back with a bounded partial inverse σ of B^{joint} on the central subalgebra. By Lemma 4.1, $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max} + 1)$ is finite and non-zero, so σ exists on the central subalgebra and is bounded by the same scale; the dual residual is symmetric to (29).

(31) is the Lipschitz-distortion height of Latrémolière [16] Def. 3.5, given by

$$\text{height}_B = \sup_{\|\nabla^{\text{joint}} f\|_\infty \leq 1} \left| \|f\|_{\text{Lip}} - \|B^{\text{joint}}(f)\|_{\text{Lip}^{\text{op}}} \right|,$$

where $\|B^{\text{joint}}(f)\|_{\text{Lip}^{\text{op}}} := \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}}$ is the Lipschitz seminorm induced by the truncated Lorentzian Dirac. By Lemma 4.8(d) and an L4 approximate-identity step on the gradient (Stein–Weiss / Paper 38 Appendix A factor-wise), the height is bounded by $C_3^{\text{joint}} \cdot \gamma^{\text{joint}}$.

(32) follows because P^{joint} is an orthogonal projection (Stinespring lift), of cb-norm 1, and introduces no Lipschitz distortion. $\square$

5.3 Main theorem

Theorem 5.3 (Lorentzian propinquity convergence on truncated $\text{SU}(2) \times U(1)_T$ Krein spectral triples). *Let $n_{\max} \geq 1$, $N_t \geq 1$, $T > 0$. The $\mathcal{K}^+$ -restricted weak-form Lorentzian Latrémolière propinquity between the truncated Krein metric spectral triple $\mathcal{T}_{n_{\max}, N_t, T}^L$ and the continuum reference triple $\mathcal{T}_{S^3 \times S_T^1}^L$ satisfies*

$$\Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{joint}}(n_{\max}, N_t) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0, \quad (33)$$

with $C_3^{\text{joint}}(n_{\max}, N_t) \leq 1$ asymptotically tight (inherited from Paper 38 / Lemma 4.4) and $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ the joint mass-concentration moment (Lemma 4.8(c)).

Proof. By Definition 2.1 and Proposition 5.2,

$$\begin{aligned} \Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) &= \Lambda(\mathcal{T}_{n_{\max}, N_t, T}^+, \mathcal{T}_{S^3 \times S_T^1}^+) \\ &\leq \max(\gamma^{\text{joint}}, \gamma^{\text{joint}}, C_3^{\text{joint}} \cdot \gamma^{\text{joint}}, 0) \\ &= C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}. \end{aligned}$$

Since $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ vanishes in the joint limit $(n_{\max}, N_t) \rightarrow (\infty, \infty)$ and $C_3^{\text{joint}} \leq 1$ asymptotically, the right-hand side $\rightarrow 0$. $\square$

Remark 5.4 (Asymptotic rate). The joint rate factors at leading order:

$$\gamma^{\text{joint}} \leq \gamma^{\text{SU}(2)} + \gamma^{U(1)} = O\left(\frac{\log n_{\max}}{n_{\max}}\right) + O\left(\frac{T}{N_t}\right),$$

with the $\text{SU}(2)$ leading constant $\lim_{n_{\max} \rightarrow \infty} n_{\max} \gamma_{n_{\max}}^{\text{SU}(2)} / \log n_{\max} = 4/\pi$ rigorous (Paper 38 § L2 Stein–Weiss closure + Paper 40 § 3.2 universal cancellation), and the $U(1)$ leading constant $\gamma_{N_t, T}^{U(1)} \leq C T \log N_t / N_t$ standard (Katznelson [15] Ch. I).

Remark 5.5 (Limit identification). The propinquity limit point is the $\mathcal{K}^+$ -restricted standard metric spectral triple $\mathcal{T}_{S^3 \times S_T^1}^+ = (P_+ C^\infty(S^3 \times S_T^1) P_+, \mathcal{K}_\infty^+, P_+ D_\infty^L P_+)$ on the $\mathcal{K}^+$ subspace of the continuum Krein space. By Latrémolière’s weak-form theorem [16] Thm. 5.5, the limit identification of Λ -convergence is on the Wasserstein–Kantorovich state space of the limit triple; this carries over to Λ^L via Definition 2.1. The weak-form caveat is that the strong-form Lorentzian propinquity (without $\mathcal{K}^+$ restriction) would identify a Krein-state-space limit, which remains open; see § 7.3.

Remark 5.6 (The proof shape is bookkeeping). In the same spirit as Paper 38 § L5 and Leimbach–vS [18], the propinquity assembly is the bookkeeping layer of the proof shape. The analytical content is supplied by Lemmas 3.1, 4.1, 4.4, 4.8; each step in Proposition 5.2 cites one of them. No new analytical input enters the assembly.

$(n_{\max}, N_t)$	$\gamma^{\text{SU}(2)}$	$\gamma^{U(1)}$	C_3^{joint}	cb-norm	Λ^L bound	$N_t = 1$ residual
(2, 1)	2.0746	$\pi/2 \approx 1.5708$	0.5774	2/3	2.0746	0.0 bit-exact
(3, 1)	1.6101	$\pi/2$	0.7071	1/2	1.6101	0.0 bit-exact
(4, 1)	1.3223	$\pi/2$	0.7746	2/5	1.3223	0.0 bit-exact
(2, 3)	2.0746	0.7220	0.5774	2/3	2.0746	—
(3, 5)	1.6101	0.4956	0.7071	1/2	1.6101	—
(4, 7)	1.3223	0.3841	0.7746	2/5	1.3223	—

Table 1: Numerical verification panel for Theorem 5.3. $\gamma^{\text{SU}(2)}$ values from Paper 38 § L2 closed-form Stein–Weiss computation; $\gamma^{U(1)}$ values from standard Fejér-on- S_T^1 at $T = 2\pi$; C_3^{joint} from the Avery harmonic saturation $(N - 1)/\sqrt{N^2 - 1}$ on the truncation. The bound on Λ^L is dominated by the SU(2) factor at every cell. The $N_t = 1$ Riemannian-limit residuals are bit-exact 0.0 in float64, the load-bearing falsifier for the construction.

6 Numerical verification and Riemannian-limit recovery

6.1 Verification panel

The driver `debug/13b_2_sub_sprint_D_compute.py` builds the joint tunneling pair on the panel

$$(n_{\max}, N_t) \in \{(2, 1), (3, 1), (4, 1), (2, 3), (3, 5), (4, 7)\}$$

at $T = 2\pi$. The panel verifies four assertions: (i) Riemannian-limit recovery at $N_t = 1$; (ii) monotone decrease of Λ^L in $n_{\max}$ at fixed N_t ; (iii) asymptotic rate consistency $\Lambda^L(4, 7)/\Lambda^L(2, 3) \approx 0.637$; (iv) constituent identities $\text{reach}_B, \text{reach}_P \leq \gamma^{\text{joint}}$ at every cell.

The numerical results are tabulated in Table 1 (reproduced from `debug/data/13b_2_sub_sprint_D.json`; see Appendix B for full data).

6.2 Riemannian-limit recovery (load-bearing falsifier)

Proposition 6.1 (Riemannian-limit recovery). *At $N_t = 1$, the $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity reduces bit-exactly to Paper 38’s SU(2) Riemannian propinquity:*

$$\Lambda^L(\mathcal{T}_{n_{\max}, 1, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) = \Lambda_{n_{\max}}^{\text{Paper 38}} \quad (\text{Frobenius residual 0.0 in float64}). \quad (34)$$

Proof. At $N_t = 1$, the only kept Fourier mode is $q = 0$, the $U(1)$ Fejér kernel collapses to the uniform Haar density $1/T$, and the temporal Berezin factor $B^{U(1)}$ is the trivial 1×1 identity. By the tensor-product factorization (27), $B_{n_{\max}, 1, T}^{\text{joint}}(f_s \otimes \mathbf{1}) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes I_1$, which equals Paper 38’s chirality-doubled spinor lift of the spatial Berezin map.

The chirality-doubled spinor lift on $\mathcal{K}^+$ is bit-identical to Paper 38’s scalar Fock-basis construction up to the basis representation: the Plancherel weights $\hat{K}^{\text{SU}(2)}(N)$, the Lipschitz bound constant $C_3^{\text{Paper 38}} \leq 1$, and the convergence rate $\gamma_{n_{\max}}^{\text{SU}(2)}$ are the same numerical objects. Therefore (34) holds. The bit-exact residual at $n_{\max} \in \{2, 3, 4\}$ is verified numerically in Table 1. $\square$

The Riemannian-limit recovery is the load-bearing falsifier of the construction: were it to fail, we would have introduced an extraneous factor in the Wick-rotated signature, miscounted Berezin weights or grids, or had a sign error in the $\mathcal{K}^+$ restriction. The bit-exact recovery rules out all three. This is the same load-bearing-falsifier pattern used in Paper 42 (Riemannian limit of the Tomita–Takesaki construction at $N_t = 1$ to Paper 38 / Paper 42 § 11) and Paper 43 (Riemannian limit of the Lorentzian Dirac at $N_t = 1$ to truthful Camporesi–Higuchi spatial Dirac).

6.3 Asymptotic rate consistency

The ratio $\Lambda^L(4, 7)/\Lambda^L(2, 3) = 1.3223/2.0746 = 0.6374$ is monotone decreasing across the panel, matching the single-factor Paper 38 and Paper 39 [37] ratios bit-identically. This confirms that the $SU(2)$ factor dominates the joint rate at the tested cutoffs, which is the expected behavior given $\gamma^{SU(2)} = O(\log n_{\max}/n_{\max})$ and $\gamma^{U(1)} = O(T/N_t)$: for $N_t \gtrsim n_{\max}/\log n_{\max}$, the $SU(2)$ factor controls the rate.

7 Discussion

7.1 Four-witness Wick-rotation connection

Paper 42 [39] closes the four-witness Wick-rotation theorem (Hartle–Hawking [12], Sewell [27], Bisognano–Wichmann [2], Unruh [31]) at the operator-system level on the *Riemannian* truncated Camporesi–Higuchi spectral triple, via the geometric BW- α and Tomita–Takesaki BW- γ modular Hamiltonians on a hemispheric wedge. Paper 43 [40] extends that closure to signature $(3, 1)$ on the Krein wedge.

The present paper supplies the Lorentzian *convergence* leg of the arc: Theorem 5.3 shows that the family of truncated Krein spectral triples on which Paper 42/43 work converges in the $\mathcal{K}^+$ -restricted weak-form propinquity to the continuum reference triple as $(n_{\max}, N_t) \rightarrow \infty$. The triple of papers (42, 43, 45) together constitute the Lorentzian-extension closure of the math.OA quartet (38, 39, 40 on the Riemannian side).

7.2 Relation to Mondino–Sämman

The synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] extends their earlier work [19] to Lorentzian pre-length spaces via ϵ -nets of causal diamonds. This is adjacent at the conceptual level (both “Lorentzian Gromov–Hausdorff-type convergence”) but operates on a categorically different mathematical object.

Remark 7.1 (Different objects, different distances). Mondino–Sämman work on pre-length spaces with causal-diamond geometry; they do not have a Dirac operator or any spectral data, and their convergence is in the synthetic-geometry metric on the space of isometry classes of such pre-length spaces. The present paper works on truncated Krein spectral triples; the convergence is in the operator-algebraic Latrémolière propinquity, restricted to the Krein-positive state-space substrate of the operator system. Neither construction implies the other; both are valid notions of Lorentzian-type Gromov–Hausdorff convergence on their respective categories.

7.3 Open questions

(Q1) **Strong-form Lorentzian propinquity.** A Latrémolière-style metric on Krein-signature spectral triples *without* the $\mathcal{K}^+$ restriction remains an open NCG-math problem. The technical issue: the standard Latrémolière construction uses the Lipschitz seminorm $\|[D, \cdot]\|_{\text{op}}$, which is well-defined on Hilbert space but requires an indefinite-inner-product analog on a Krein space with appropriate Krein-self-adjointness on D_L per vdD / BBB. The published math.OA literature as of May 2026 does not provide this construction. By performing the $\mathcal{K}^+$ restriction we trade the strong-form construction for one that uses Hilbert-space machinery on $\mathcal{K}^+$ and obtains a defensible weak-form convergence theorem at the $\mathcal{K}^+$ -state-space / Wasserstein–Kantorovich level. Extending to strong form is a multi-month original NCG-math project, and the structural sharpening of Paper 42 § 10 question O3 (the spectral-action-vs-modular-Hamiltonian distinction) suggests the strong form will need additional structural input.

- (Q2) **De-compactification limit** $T \rightarrow \infty$. The present construction is at compact temporal radius T . The $T \rightarrow \infty$ limit recovers the non-compact $\mathbb{R}_t$ direction; the joint mass-concentration moment $\gamma^{U(1)} = O(T/N_t)$ then requires N_t to grow at least as fast as T , and the asymptotic rate analysis must be re-examined in this regime. This is internally a separate sprint (L3c in our taxonomy) and is not addressed here.
- (Q3) **Cross-manifold extensions**. The cross-manifold extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ is blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (Paper 24 [34] § V). The four-layer Coulomb/HO asymmetry there (spectrum-computing L_0 , calibration π , Wilson gauge with natural matter, modular-Hamiltonian structure of the wedge KMS state) precludes the natural extension within the present framework.
- (Q4) **Sharp asymptotic-tight** $C_3^{\text{joint}} = 1$. The present bound has $C_3^{\text{joint}} \leq 1$ asymptotically tight, with finite-cutoff bound $\sup_{N \leq n_{\max}} (N-1)/\sqrt{N^2-1} < 1$. A direct proof of $C_3^{\text{joint}} = 1$ in the joint limit (rather than inheriting from the $SU(2)$ factor) would require a finer joint analysis of the temporal direction's contribution to the Lipschitz seminorm when restricted to non-pure-tensor multipliers; this is not needed for the qualitative-rate convergence theorem here.
- (Q5) **Quantitative rate constants on the $U(1)$ factor**. The $U(1)$ leading constant in $\gamma^{U(1)} = O(T \log N_t/N_t)$ is standard Cesàro/Fejér; a Stein–Weiss-style derivation of the joint constant matching the $SU(2)$ factor's $4/\pi$ would be a useful refinement, but is not needed for the qualitative-rate convergence theorem.

7.4 Structural-skeleton scope

The construction obeys the structural-skeleton scope discipline of the GeoVac framework: the convergence theorem determines the rate, the asymptotic constant, and the limit identification on the $\mathcal{K}^+$ -substrate at finite cutoff. It does not determine calibration data (Yukawa selection, inner-factor parameters), which are external input. The selection of $T = 2\pi$ is structurally tied to the Bisognano–Wichmann modular period via Paper 43; the construction itself works for any $T > 0$.

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A Symbolic Plancherel calculation for the joint Fejér symbol

We record the symbolic factorization of the joint Plancherel symbol $K^{\hat{\text{joint}}}(j, k)$ used in Lemma 4.1.

Lemma A.1. *For all $n_{\max} \geq 1$, $N_t \geq 1$, the joint Plancherel symbol factorizes exactly in $\mathbb{Q}$:*

$$K^{\hat{\text{joint}}}_{n_{\max}, N_t}(j, k) = \hat{K}^{\text{SU}(2)}_{n_{\max}}(j) \cdot \hat{K}^{U(1)}_{N_t}(k),$$

with

$$\begin{aligned} \hat{K}^{\text{SU}(2)}_{n_{\max}}(j) &= \frac{2j+1}{Z^{\text{SU}(2)}_{n_{\max}}} \mathbf{1}_{\{j \leq j_{\max}\}}, & Z^{\text{SU}(2)}_{n_{\max}} &= \frac{n_{\max}(n_{\max}+1)}{2}, \\ \hat{K}^{U(1)}_{N_t}(k) &= \max\left(0, \frac{N_t+1-2|k|}{N_t+1}\right). \end{aligned}$$

Proof. For the $SU(2)$ factor, the spectral Fejér kernel $K^{SU(2)} = Z^{-1} |\sum_{j \leq j_{\max}} \sqrt{2j+1} \chi_j|^2$ has Plancherel symbol computed by orthogonality of characters on $SU(2)$: $\int_{SU(2)} K^{SU(2)} \chi_{j'} dg = (2j'+1)/Z$ for $j' \leq j_{\max}$, zero otherwise (Paper 38 § L2(a)).

For the $U(1)$ factor, the Fejér kernel $K^{U(1)} = N_t^{-1} |D_{N_t}|^2$ where $D_{N_t} = \sum_{|k| \leq K_{\max}} e^{i2\pi k\theta/T}$ is the Dirichlet kernel. Direct geometric-series computation gives $K^{\hat{U}(1)}(k) = \max(0, (N_t + 1 - 2|k|)/(N_t + 1))$ (Katznelson [15] Ch. I.2).

The joint kernel is the tensor product on the direct product group; the joint Plancherel symbol on the dual $\hat{S}U(2) \times \hat{U}(1)$ factorizes correspondingly via Peter–Weyl on $SU(2) \times U(1)$ (direct product gives external tensor products of irreps; characters multiply). $\square$

B Numerical verification panel data

The full numerical panel data tabulated in Table 1 is reproduced from `debug/data/l3b_2_sub_sprint_D.json`. The corresponding computational driver is `debug/l3b_2_sub_sprint_D_compute.py`, and the production module is `geovac/lorentzian_propinquity_compact_temporal.py` (approximately 700 lines, with the `LorentzianTunnelingPair` dataclass, the `LorentzianPropinquityBound` dataclass, the `compute_lorentzian_propinquity_bound` function, and the `verify_riemannian_limit_at_N_t` bit-exact check).

The numerical verification of all constituents in Proposition 5.2 is in `tests/test_lorentzian_propinquity`. (15+ test functions covering the tunneling pair construction, reach and height bounds, the main theorem panel, the asymptotic rate consistency, the $\mathcal{K}^+$ restriction correctness, the Riemannian-limit recovery, and the monotone decrease in $n_{\max}$).

Convergence diagnostic. The ratio $\Lambda^L(4, 7)/\Lambda^L(2, 3) = 1.3223/2.0746 = 0.6374$ matches the single-factor Paper 38 / Paper 39 ratio bit-identically, confirming that the $SU(2)$ factor dominates the joint rate at the tested cutoffs. At larger cutoffs the $U(1)$ factor would re-enter at order $O(T/N_t)$ if N_t does not grow as fast as $n_{\max}/\log n_{\max}$.

Memory note. The panel cell $(n_{\max}, N_t) = (5, 9)$ was originally planned but exceeds the operator-system vec-matrix memory budget at $\dim_K = 1260$ and $K = 2565$ multipliers (≈ 60 GiB `complex128`). The panel is restricted to $\{(2, 3), (3, 5), (4, 7)\}$, which is sufficient for Riemannian-limit recovery, monotone-decrease, and asymptotic-rate consistency. Larger cells would require sparse linear-algebra refactoring of `geovac.operator_system_compact_temporal`, which is a downstream engineering item and not an analytical blocker.

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